



Tutorials ▾

References ▾

Exercises ▾

Certificates ▾

Search



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[HTML](#) [CSS](#) [JAVASCRIPT](#) [SQL](#) [PYTHON](#) [JAVA](#) [PHP](#) [HOW TO](#) [W3.CSS](#) [C](#) [C++](#) [C#](#) [BOOT! >](#)

NumPy Tutorial

[NumPy HOME](#)[NumPy Intro](#)[NumPy Getting Started](#)[NumPy Creating Arrays](#)[NumPy Array Indexing](#)[NumPy Array Slicing](#)[NumPy Data Types](#)[NumPy Copy vs View](#)[NumPy Array Shape](#)[NumPy Array Reshape](#)[NumPy Array Iterating](#)[NumPy Array Join](#)[NumPy Array Split](#)[NumPy Array Search](#)[NumPy Array Sort](#)[NumPy Array Filter](#)

NumPy Random

[Random Intro](#)[Data Distribution](#)[Random Permutation](#)[Seaborn Module](#)[Normal Distribution](#)[Binomial Distribution](#)[Poisson Distribution](#)[Uniform Distribution](#)[Logistic Distribution](#)[Multinomial Distribution](#)[Exponential Distribution](#)[Chi Square Distribution](#)[Rayleigh Distribution](#)[Pareto Distribution](#)

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NumPy Array Copy vs View

[◀ Previous](#)[Next ▶](#)

The Difference Between Copy and View

The main difference between a copy and a view of an array is that the copy is a new array, and the view is just a view of the original array.

The copy *owns* the data and any changes made to the copy will not affect original array, and any changes made to the original array will not affect the copy.

The view *does not own* the data and any changes made to the view will affect the original array, and any changes made to the original array will affect the view.

COPY:

Example

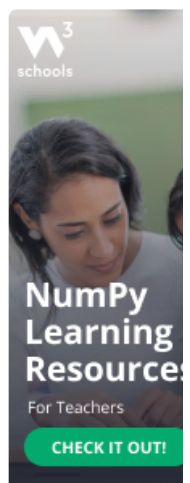
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Make a copy, change the original array, and display both arrays:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])
x = arr.copy()
arr[0] = 42

print(arr)
print(x)
```

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AD

The copy SHOULD NOT be affected by the changes made to the original array.

VIEW:

Example

Make a view, change the original array, and display both arrays:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])
x = arr.view()
arr[0] = 42

print(arr)
print(x)
```

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The view SHOULD be affected by the changes made to the original array.

Make Changes in the VIEW:

Example

Make a view, change the view, and display both arrays:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])
x = arr.view()
x[0] = 31

print(arr)
print(x)
```

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The original array SHOULD be affected by the changes made to the view.

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Check if Array Owns its Data

As mentioned above, copies *owns* the data, and views *does not own* the data, but how can we check this?

Every NumPy array has the attribute `base` that returns `None` if the array owns the data.

Otherwise, the `base` attribute refers to the original object.

Example

Print the value of the base attribute to check if an array owns its data or not:

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5])

x = arr.copy()
y = arr.view()

print(x.base)
print(y.base)
```

[Try it Yourself »](#)

The copy returns `None`.